Higher National Diploma in Information Technology
First Year, Second Semester Examination – 2023
HNDIT2052 Principles of User Interface Design

Instructions for Candidates:
Answer any five (05) questions.
Every question carries 20 marks

No. of questions : 06
No. of pages : 03
Time : Three (03) hours

Question 01.

(i). What are the components of Human-Computer Interaction? Explain each of them. [04 Marks]

(ii). Explain the importance of Human-Computer Interaction (HCI) in modern software development. [04 Marks]

(iii). What is an *Interactive system*? Give two (02) examples for interactive systems. [04 Marks]

(iv). Compare and contrast the skills of an interactive systems designer with those of a traditional software programmer. [04 Marks]

(v). "Understanding the user and their context is the cornerstone of effective UI design." Do you agree with this statement? Justify your answer. [04 Marks]

[20 Marks]

Question 02.

(i). What is *multimodal interaction*? List two (02) advantages of multimodal interaction. [04 Marks]

(ii). Explain the concept of "*two gulfs in the interaction*" and how a good design can minimize the gulfs. [04 marks]

(iii). What does it mean by "*Ergonomics*" in design? List two (02) advantages of ergonomic user interfaces. [04 marks]

(iv). List two (02) common interaction styles or paradigms and provide a brief advantage of each. [04 marks]

(v). Draw a low-fidelity wireframe for the **login** screen. The **login** screen allows users to enter their username and password, and also provides access to the 'New User Registration' form.

[04 Marks]

[20 Marks]

Question 03.

(i). Describe the followings types of typefaces with its best suit context. [04 Marks]

(a). Serif font

(b). Sans-Serif font

(c). Monospaced

(d). Display font

(ii). List two (02) factors affecting readability and two (02) of common mistakes in UI typography.

[04 Marks]

(iii). Choose the right answer for the blanks from the given list:

(a). The _____ color model is primarily used for light-emitting devices like screens. (***CMYK, RGB***) [02 Marks]

(b). Among the three main components of color, (**hue, saturation, and brightness**) _____ refers to the actual color perceived by the human eye. [02 Marks]

(iv). Identify the color palette described in each given context below:

(a). A color scheme created using different shades, tints, and tones of a single base color.

[01 Mark]

(b). A color scheme that uses three colors that are evenly spaced around the color wheel.

[01 Mark]

(c). A color scheme that consists of colors located directly adjacent to each other on the color wheel. [01 Mark]

(d). A color scheme formed by combining two adjacent colors on one side of the color wheel with the color directly opposite the selected color. [01 Mark]

(v). "Effective use of typography and color theory can significantly impact the user's perception and interaction with an interface." Do you agree with this statement? Justify your answer.

[04 Marks]

[20 Marks]

Question 04.

(i). Describe the *PACT framework* for analyzing interactive systems, outlining what each letter in PACT represents. [04 Marks]

(ii). *People* are the users of system. Explain how they are differed from each other based on their characteristics. [04 Marks]

(iii). Mention four (04) benefits of adopting a Human-Centered Design (HCD) approach in the interface design process. [04 Marks]

(iv). "Offer error prevention and simple error handling" and "Offer informative feedback" is two of golden rules to be applied in UID. Explain them with one example for each. [04 marks]

(v). "While design principles offer valuable guidelines, strict adherence to them always guarantees a user-centered design." Do you agree with this statement? Justify your answer. [04 marks]

[20 Marks]

Question 05.

(i). Define the terms *Usability* and *Accessibility*. [04 Marks]

(ii). List 04 dimensions of usability. [04 Marks]

(iii). What are the primary stages in the User Interface Design process? [04 Marks]

(iv). What is *Hierarchical Task Analysis*? List two (02) advantages of it in UID. [04 Marks]

(v). "Iteration is key to successful GUI design." Do you agree with the statement? Justify your answer. [04 Marks]

[20 Marks]

Question 06.

(i). What is a *prototype*? List two (02) advantages of a prototype in UID. [04 Marks]

(ii). Differentiate between "low-fidelity prototype" and "high-fidelity prototype". [04 Marks]

(iii). List two (02) software tools for high- fidelity prototyping and two (02) factors to be consider selecting a prototyping tool for your UID project. [04 Marks]

(iv). List four (04) issues in user interface designing. Explain 02 of them. [04 Marks]

(v). How does prototyping help in identifying user requirements and improving system usability? [04 Marks]

[20 **Marks]**